A Comprehensive Resource of Anti-MRSA Peptides and Essential Oils

Rajat Kumar Mondal[1,2], Sintu Kumar Samanta[1]

[1]Biochemistry and Bioinformatics Laboratory, Department of Applied Sciences, Indian Institute of Information Technology, Allahabad, Devghat, Jhalwa, Prayagraj, Uttar Pradesh, 211012, India; [2]GeneArche Wellness Pvt. Ltd., Pune, Maharashtra, 411005, India

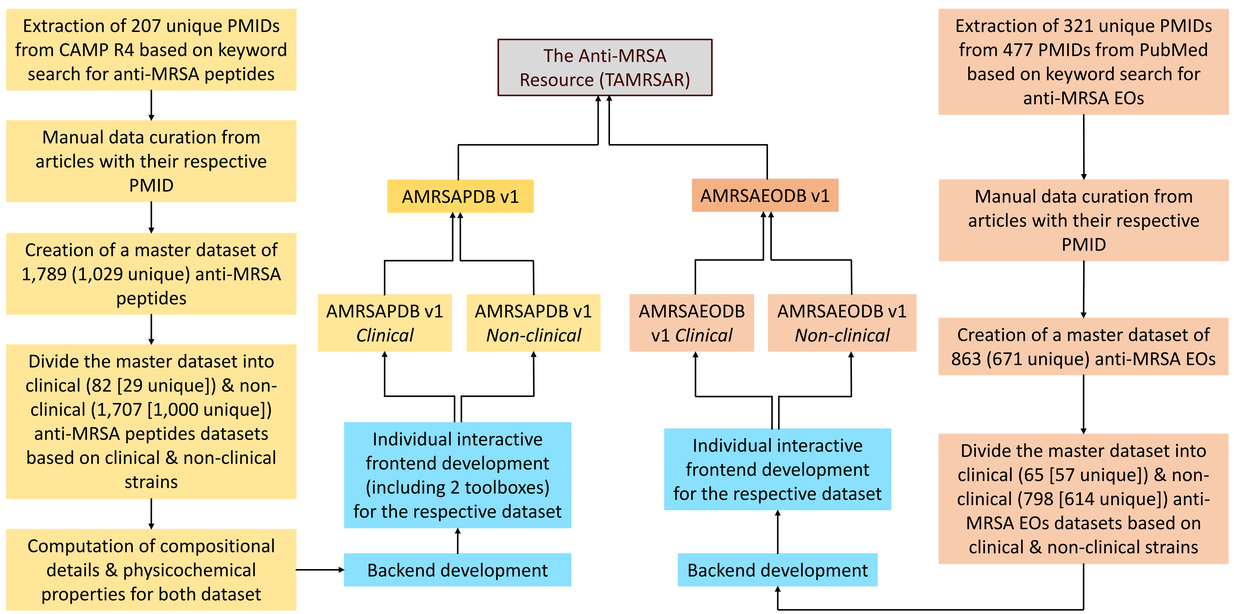
Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) is a major antimicrobial-resistant pathogen, with infection cases rising from 21% in 2016 to 35% in 2020. This increase has driven research into alternative treatments like antimicrobial peptides (AMPs) and essential oils (EOs), which have shown higher success rates than other options. To support this, we developed The Anti-MRSA Resource (TAMRSAR), a database compiling 1,789 AMPs (1,029 unique) and 863 EOs (671 unique) effective against MRSA. Data was manually extracted from PubMed literature, with integrated tools for sequence analysis and feature assessment. TAMRSAR is freely accessible at https://bblserver.org.in/tamrsar/.

Introduction

- MRSA infections surged by 14% between 2016 and 2020, with colonization in adults aged 40–85 increasing the risk of death by 75% (Hazard Ratio = 1.75).
- MRSA is resistant to almost all Beta-lactam antibiotics, and even recommended treatments like vancomycin and daptomycin suffer from low bactericidal activity, rising resistance, and frequent treatment failures.
- According to the WHO BPPL 2024, MRSA is categorized as a high-priority pathogen that poses a significant global threat to human health.
- AMPs and EOs are highly effective against MRSA due to their ability to target multiple bacterial pathways, including membrane disruption and intracellular mechanisms, making it harder for MRSA to develop resistance compared to conventional antibiotics.
- In the study, we curated 1,789 AMPs (1,029 unique) and 863 EOs (671 unique) effective against MRSA and developed TAMRSAR, an open-access resource with 4 child databases and 8 interactive tools (inside 2 toolboxes).

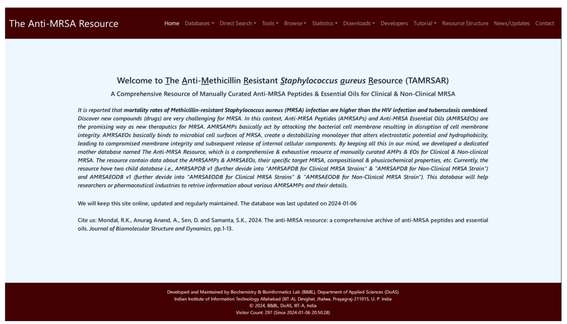
Methodology



AMRSAPDB v1 → Anti-Methicillin Resistant *S. aureus* Peptide Database version 1; AMRSAEODB v1 → Anti-Methicillin Resistant *S. aureus* Essential Oils Database version 1

Results

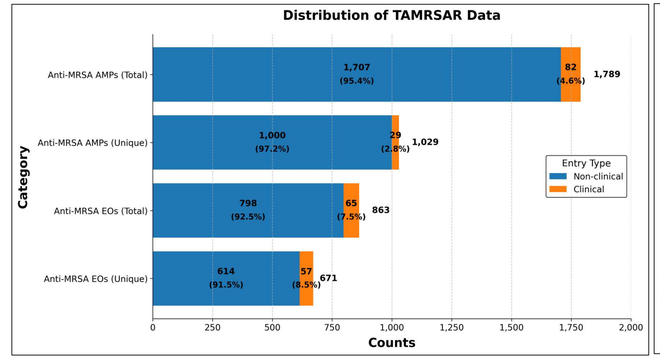
TAMRSAR

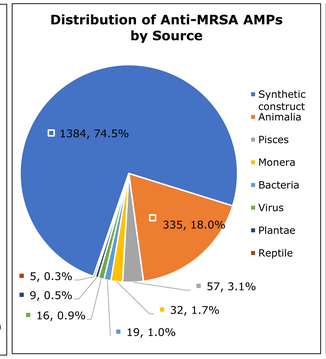


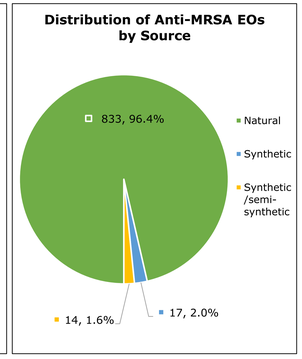
Key Findings

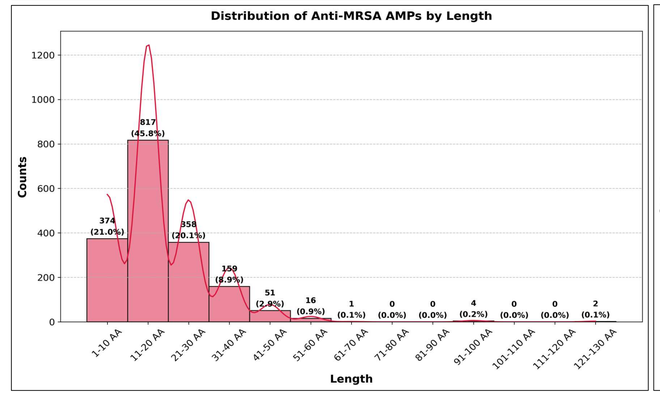
- 1789 anti-MRSA peptides (1029 unique)
- 863 essential oils (671 unique) targeting MRSA
- Clinical vs. Non-clinical Data: 82 peptides & 65 EOs validated against clinical MRSA strains. Others are against non-clinical strains of MRSA
- Peptide Length Distribution: 12-50 amino acids
- Data Retrieval Feature: Basic text-based search, Advanced text-based search
- Data Export: Multiple formats (FASTA [not applicable for EO entries], TSV, Custom TSV, CSV, JSON, etc.)
- Tools: 8 tools (inside 2 toolboxes) for sequence alignment (like BLASTp, MUSCLE), and peptide feature computation (like physicochemical properties, QSAR, CTD)

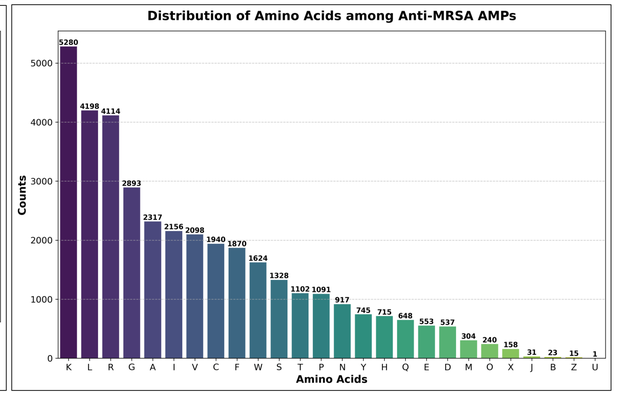
Data Analysis

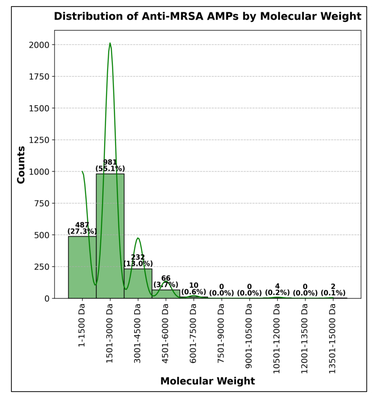


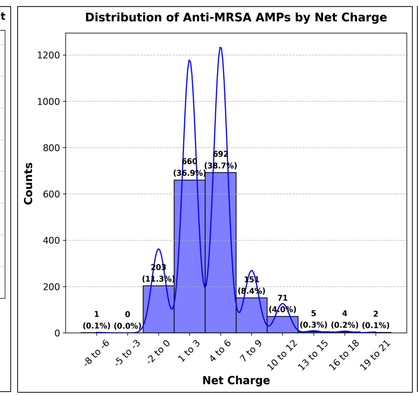


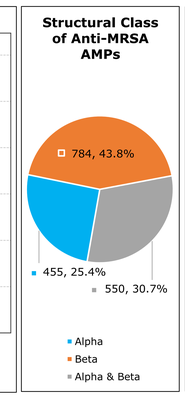


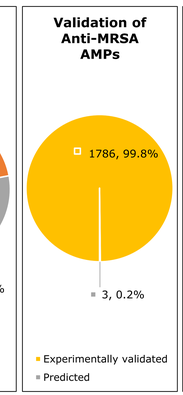


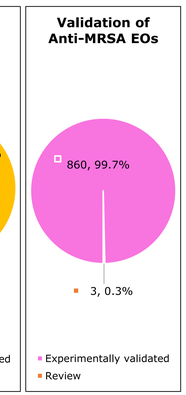












Discussion & Conclusion

- TAMRSAR is a comprehensive, dedicated resource featuring anti-MRSA peptides and essential oils (EOs), along with clinical and non-clinical differentiation, filling a crucial gap left by ANTISTAPHYBASE.
- The resource is designed to aid in overcoming the challenges of antibiotic resistance in MRSA by facilitating the identification and optimization of potential anti-MRSA agents.

Acknowledgments

RKM is thankful to IIIT-A for the infrastructural facilities and to MoE-GoI for the MTech fellowship during this project.

Future Work

- Incorporation of anti-MRSA peptide from sources like APD3, DRAMP, and EOs from databases like EssOilDB and the EOU Database.
- A cutting-edge prediction tool for identifying anti-MRSA peptides.
- Addition of extensive data annotations like mode of action (MoA) and toxicity details for each entry.
- A more modern UI and APIs for seamless data access.

Reference

Mondal, R.K., Anurag Anand, A., Sen, D. and Samanta, S.K., 2024. The anti-MRSA resource: a comprehensive archive of anti-MRSA peptides and essential oils. *Journal of Biomolecular Structure and Dynamics*, pp.1-13.